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Beyond Traditional Methods: RFID Technology-Based Solutions for Upper Completions

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Abstract

Traditional methods for isolating upper completion strings often involve complex, risk-prone, and costly intervention operations. This paper introduces a pioneering approach that leverages radio frequency identification (RFID) technology to eliminate the need for such interventions or conventional ball-drop systems. This paper focuses on the adoption of an intervention-less radio frequency identification (RFID) reservoir isolation valve (RIV) for upper completion installations.

Traditional methods for isolating, such as ball and seat, dart, and e-line conveyed intervention plugs, often introduce complexities, risks, and significant costs. These methods can lead to increased intervention time and technical challenges, particularly when circulating balls or darts onto seats. To address these limitations, an innovative RFID remotely activated reservoir isolation valve (RIV) was developed. This technology offers multiple built-in downhole contingencies, providing a more reliable and efficient solution for upper completion isolation. Following a thorough review of the operation requirements along with the associated historical challenges compared to conventional isolation methods, the intervention-less RFID RIV solution was developed, manufactured, tested, and subsequently deployed in a horizontal well. The technology is installed in the open position and is activated by specifically engineered RFID tags encased in a transportation carrier which allows it to be introduced to the flow stream and pumped to the toe of the well, even in highly deviated/horizontal reservoir sections. Once the RFID tag has reached the onboard electronics in the RIV, it passes a pre-programmed command to the RIV instructing it to close after a pre-determined time interval and isolate the upper completion. The programming logic also allows for built-in "timer" contingencies in the unlikely event that the RFID tags do not reach the tool as intended.

The RFID-activated RIV was successfully deployed in a challenging wellbore environment. The valve was activated on-demand by circulating the RFID tag to its designated location, ensuring precise and timely isolation of the upper completion. Subsequent operations, such as setting production packer, were carried out against the securely isolated upper completion string, enabled by the RFID RIV. This streamlined workflow eliminated the need for multiple intervention runs, significantly reducing operational risk and cost.

The RFID technology yielded several key benefits as enhanced wellbore integrity, improved operational efficiency, reduced risk, and increased flexibility. The RFID RIV provided a reliable and durable barrier, preventing fluid migration and ensuring wellbore integrity. The streamlined workflow and reduced

intervention time led to significant cost savings and accelerated project timelines. Additionally, eliminating the need for complex and risky intervention operations minimized the potential for accidents and equipment damage. The RFID RIV's adaptability to various well configurations and operational scenarios further enhanced its flexibility and utility.

The successful deployment of the RFID-activated demonstrates the potential of innovative technologies to improve the efficiency, safety, and cost-effectiveness of well completion operations. By adopting this technology, the engineering teams can significantly enhance the efficiency, safety, and cost-effectiveness of upper completion operations.

Introduction and The History of Technology

RFID is not a new technology. It started back in the 1940s during World War II. European countries used this technology to identify whether the aircraft were friends or enemies. This was started by using the transponders, and step by step, especially after the initial patent in 1973, it became a familiar technology and was incorporated into daily use. Figure 1 shows the RFID applications in the past versus recent applications for the technology. The RFID technology is used in various needs (Adan, 2013) as shown in Figure 1:

- Managing the supply chain through goods monitoring
- In inventory management, by automating inventory counts and reducing manual errors.
- Also, in access control, securing buildings and areas using RFID cards grants access.
- Animal tracking and identification, by inserting a tag under the animal's skin.
- Highway toll tags.
- Electronic Keys
- And many other applications

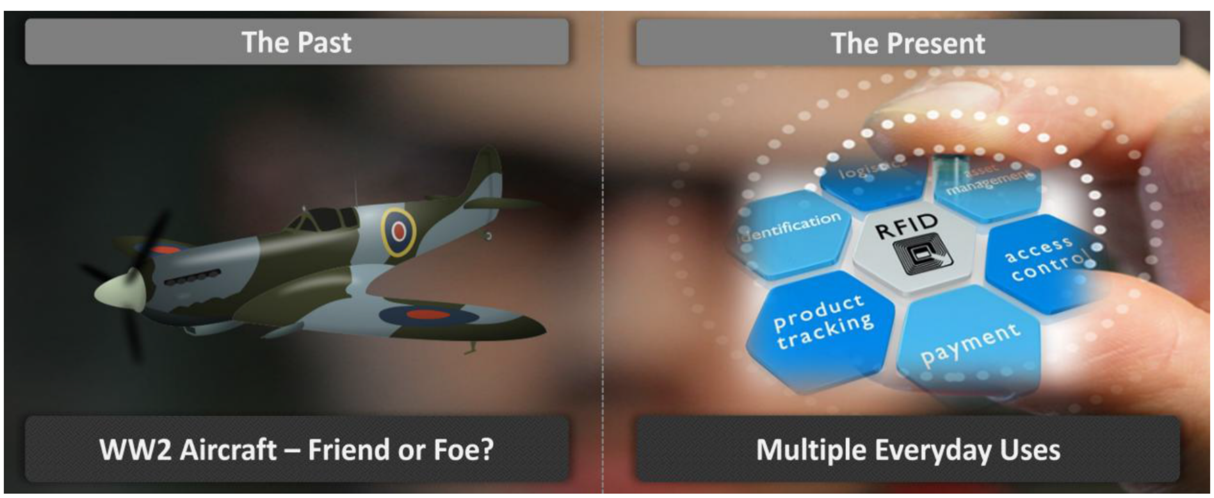

Figure 1—RFID Usage in Past and Present

RFID in the Petroleum Industry

The RFID technology is the most cutting-edge technology in the oil & gas market, since it started in 2008 by combining it with the drilling circulating sub for a specific application in the North Sea. Then over the years, it has been used with the under-gauge reamer, lower completion equipment such as frac sleeves and Inflow Control Device (ICD), and the reservoir isolation valve.

This technology tackled many challenges and mitigated risks by remotely communicating with a series of downhole completion equipment. The technology reduces operational time, costs, minimizes risks, and enhances the flexibility of operations. This can happen by eliminating the use of the Wash-pipes / Slickline / Coiled Tubing (CT) (briefly, 100% Intervention-free). It also eliminates the risk of the Drop-ball traditional method, which sometimes doesn't land on the ball seat. It gives you the ability to control the well remotely. It saves up to 60% of the Operational time, depending on the type of application. Not to mention, having the minimum equipment on the rig site minimizes the hazards, and last but not least, minimizes the people on board (Murdoch et al., 2016).

How The Technology Works

Any RFID tool in the oil & gas industry has three main components:

1. **The Antenna:** It has its internal power source. It is responsible for transmitting the electro-magnetic signals, powers up the passive tags when they are in the range of 6 to 12 inches, reads the stored instructions, and sends them to the next component, which is the electronic module. Antenna should be programmed to respond only to the tags with ID numbers and not to any other passing tags without these numbers (Figure 2).
2. **The Electronic Module:** This is the brain of the tool that contains the circuit boards and lithium batteries. Its function is to receive the instructions from the Antenna and send them to the mechanical module to activate it (Figure 3).
3. **Mechanic Module:** such as Sleeves, Flapper or Ball Valves

   Flapper valve – Intervention-less solution used to eliminate the time and cost associated with slick-line runs to isolate reservoirs or intervals. It can also be used where it is challenging to isolate the toe of the wheel, as some geometries can be challenging for landing balls on the seat (Vázquez and Murdoch, 2017). Typically, RIH opened, Tag to close and perform hydraulic events, followed by a pressure pulse to open, re-establish communication to the reservoir (Figure 4).

   Sleeves – Come in many forms depending on the application. These can be single-shot with one or two valve block assemblies or multi-cycle. They can be used to cement, stimulate, or simply flow through to exchange fluids or permit access to the reservoir with the option of combining with inflow control devices (Figure 5).

   Ball Valve – We have run several of these for various clients, allowing the ability to remotely function the ball valve from open to closed position or vice versa, potentially saving significant time and cost, as demonstrated in the upcoming sections (Figure 6).

Figure 2—The Antenna

Figure 3—The Electronic Module

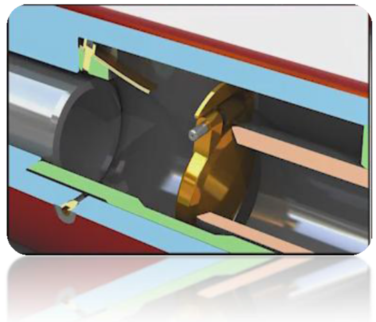

Figure 4—Flapper Valve

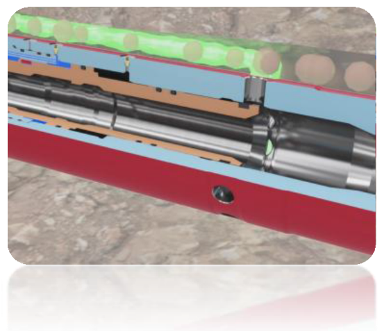

Figure 5—Sleeves

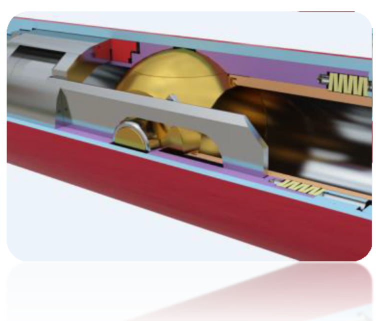

Figure 6—Ball Valve

The Communication Methods

There are three communication methods to activate any RFID tools. The RFID tool can include a single method of communication or dual or multiple (RFID-Enabled Technologies, 2025).

1. **RFID tags:** These must be programmed before running in the hole with a special code containing variables. These variables help every downhole tool to be addressed separately or collectively. The programmed command has the ability to actuate the tool immediately or after meeting specific conditions. The RFID tag is a small piece of 3.85mm in diameter and 23.1mm long (Figure 7).

   RFID tags are housed in rubber carriers. These carriers, shown in, they do not have to land or sit in the RIV, and there is no constraint as to their geometry (Edwards et al., 2017). They can therefore be exclusively designed and engineered to be carried in fluid flow whilst optimizing orientation and providing impact protection. The result of this design is that the tag carrier is much smaller and lighter compared to a conventional ball, and it travels at close to 100% of the fluid flow rate. Due to the tag carrier's soft material properties, they can pass through restrictions much smaller than their actual size. The tag carrier has a 1" OD head (Figure 8).

2. **Pressure Pulse:** Where a tag cannot be conveyed, such as in a closed system where circulation is impossible, we can apply a Pressure Pulse from the surface (Figure 9 and 10).
3. **Timer:** RFID tools can have an on-board clock, which enhances flexibility and permits the tools to function after a given delay following instructions (Figure 11).

Figure 7—RFID Tag

Figure 8—Tag Carrier

Figure 9—Pressure Pulse

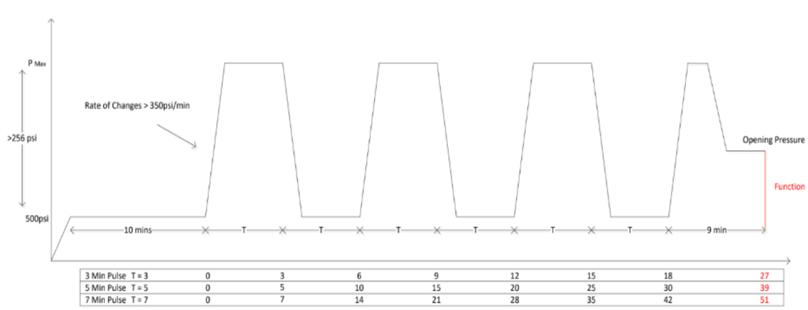

Figure 10—Pressure Pulse Graph Sample

Figure 11—Timer

Upper Completion System Configuration

In the standard upper completion trips, intervention operation is a must. Whether it's e-line, coiled tubing, or slick-line, such: operations carry many risks and time waste, ultimately leading the operator to undesirable outcomes.

Looking at the functionality offered by the RFID completion, if the operation were completed using traditional methods, several interventions would be required, such as:

- Plug Installation
- Prong Installation
- Sliding sleeve Opening TRIP.
- Sliding sleeve Closing TRIP.
- Prong Recovery
- Plug Recovery

The RFID Upper completion system comprises a simple configuration with only a couple of RFID tools (RIV and Circulating sleeve). The rest of the equipment is a standard production packer with the PBR, tubing, and float shoe. The liner hanger system is eliminated in this configuration (Figure 12).

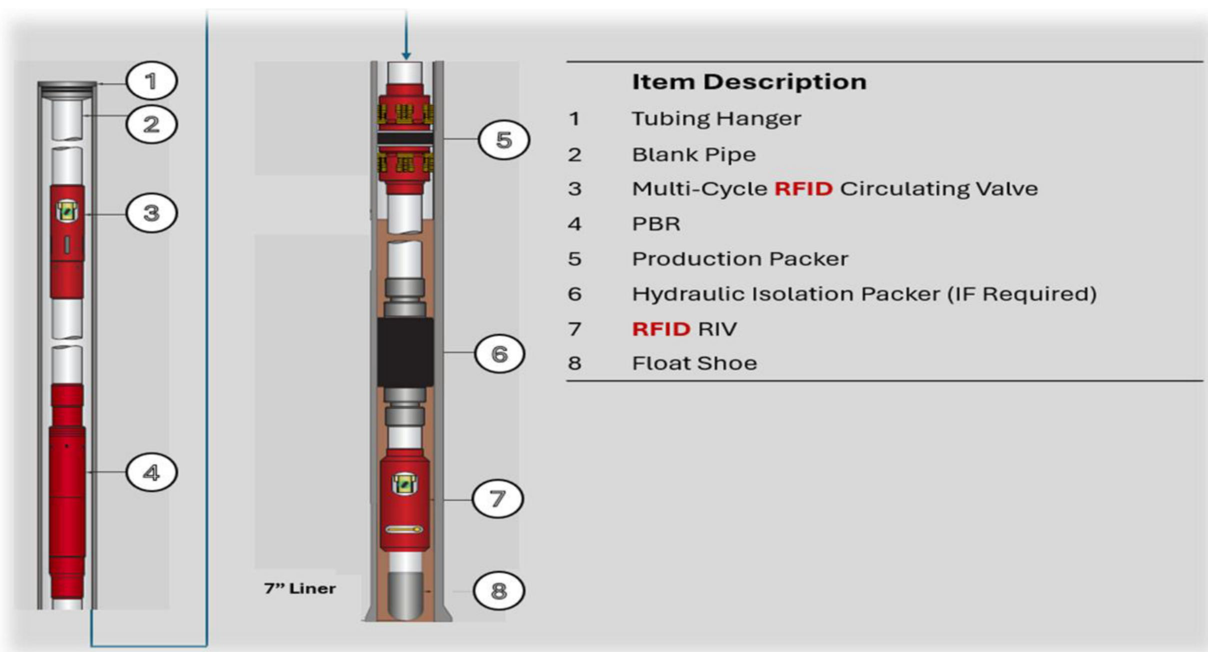

Figure 12—RFID Upper Completion System Configuration

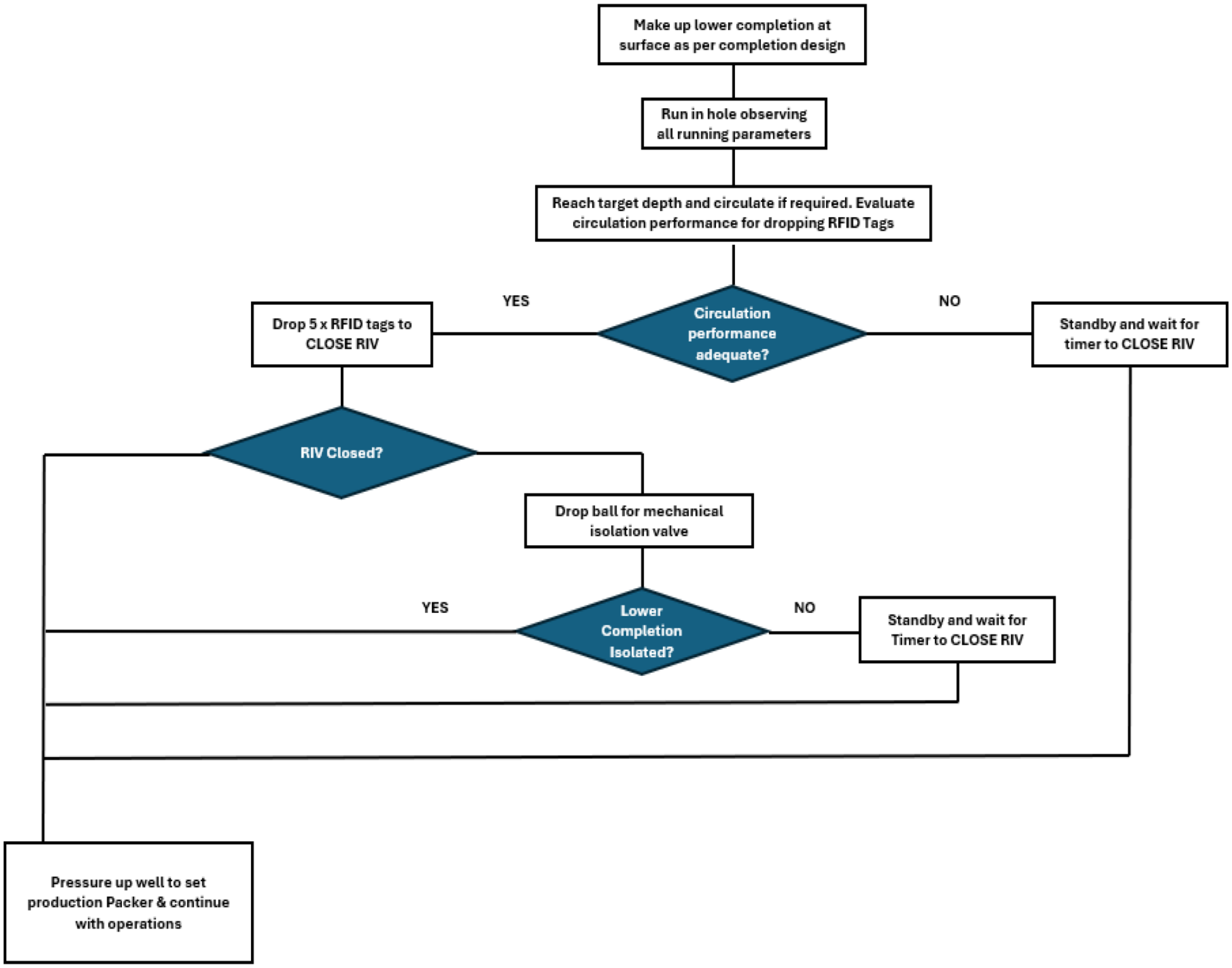

Figure 13—RFID System Flow Chart (Munro and Zaki, 2024)

The RFID RIV has been used in many applications to help activate the lower completion equipment, such as the ICDs, hydraulic open hole packers, and Frac Sleeves. In this article, we will elaborate on using the same tool in the upper completion application, shut off a barrier valve to pressure test the tubing, set the production packer, and open & close a sliding sleeve to circulate the completion fluid.

Completing all the above operational steps without any kind of intervention is impossible unless we use the RFID technology.

RFID System – Brief Running Procedures

- RIH with the upper completion string to the desired depth.
- Circulate if required.
- Once reach TD, drop the RFID tags and circulate behind them until they reach the RFID RIV to activate it in the CLOSED position.
- Confirm the RIV closure by applying 1,000psi.
- Increase the pressure to 4,000psi to set the production packer.
- Perform a pressure test to confirm the sealing integrity and perform a Push-Pull test to confirm the slip's functionality. Bleed off.

- Overpull to shear the seal assembly and disconnect it from the PBR.
- Use the pressure pulse activation method to open the RFID sliding sleeve and circulate the production fluid.
- After finishing the circulation, drop the RFID tags to close the RFID sliding sleeve.
- Space out and land the TBG Hanger.
- POOH.

RFID RIV Operational Flow Chart

RFID RIV (Reservoir Isolation Valve)

The RFID RIV is a flapper barrier system that comprises a hinged flapper that can rotate through 90˚. It is held back during the run-in-hole phase by means of the main flow tube. Once this flow tube has been shifted, the flapper can fall down and land on the sealing face. This then isolates the well, and pressure testing can commence (Figure 14).

Figure 14—RFID RIV

Certified under ISO 28781, the RIV offers a cost-efficient and reliable method for isolating completion systems. Its flapper acts as a bi-directional seal, enabling testing of the completion system and activating hydraulically set tools (Figure 15).

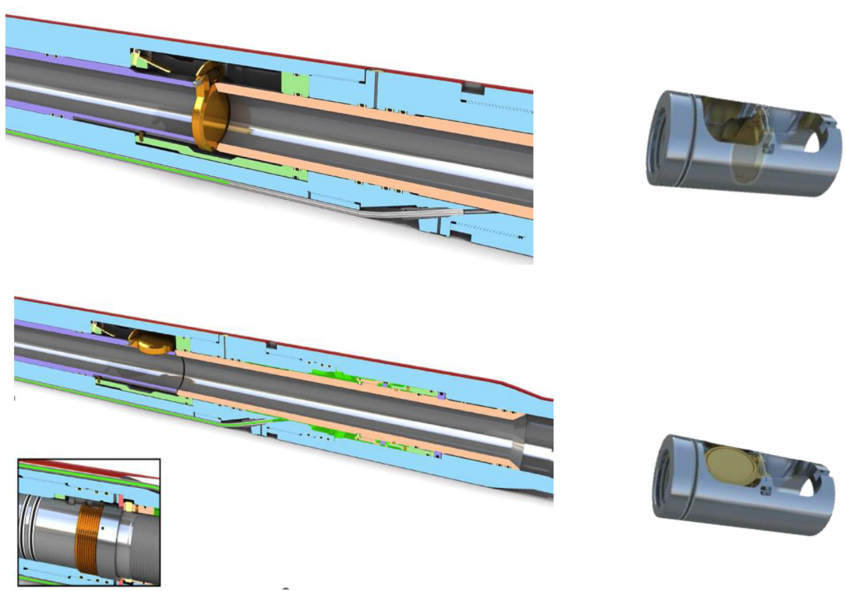

Figure 15—RIV Flapper Barrier Assembly

The RFID RIV can be provided in 2 different versions (Single cycle or Dual cycle). Each cycle requires a separate fuse (Figure 16). When the tool receives the correct trigger command via RFID tags, pressure pulse, or timer, the electronics actuate the control valve by severing a retaining cord. This allows hydrostatic well pressure to drive the flow tube piston up and release the barrier flapper.

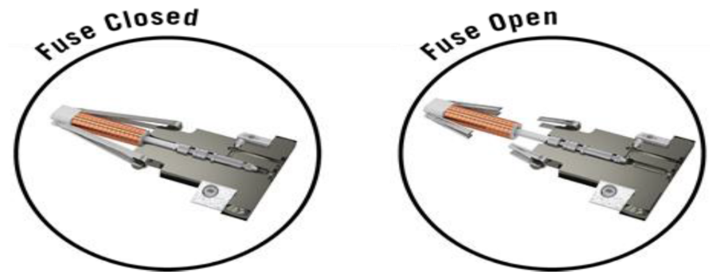

Figure 16—RIV Fuse Valve Assembly

The RFID RIV tool offers operational flexibility with both single and dual-cycle configurations, depending on the application. In a single-cycle operation, the tool is run into the wellbore in the open position to allow for circulation until the desired depth is reached. RFID tags are then deployed to activate and close the tool (Figure 17). As a contingency, if the RFID tags fail to reach the tool downhole for any reason, an internal timer serves as a backup activation method.

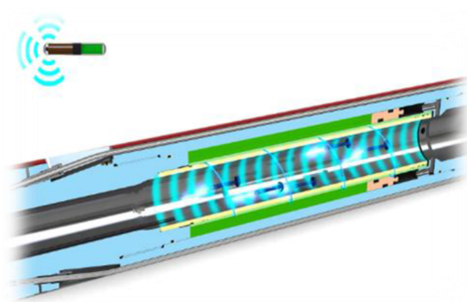

Figure 17—Drop RFID Tags

For the applications necessitating a dual-cycle operation, the initial step involves burning the first fuse, which actuates the closure of the flapper. A pressure pulse method is employed to reopen the flapper (Figure 18). This alternative method is necessary because, with the flapper closed, a flow path for circulating RFID tags to the tool is no longer available. Opening the flapper will burn the second fuse.

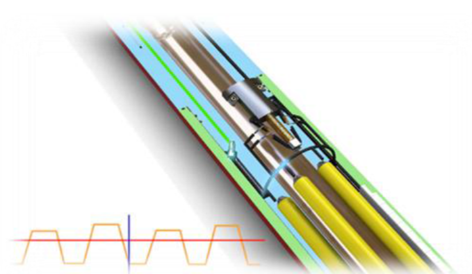

Figure 18—Pressure Pulse Activation Method

RIFD RIV Technical Specs

RFID Circulation Sleeve

The RFID remotely sliding sleeve leverages a combination of well sliding-sleeve technology and RFID capabilities to create a device for intervention-less, control-line-free well management (Figure 19). It provides remote control overflow from individual production zones, and importantly, an unlimited quantity of these sleeves that can be installed within a single mono-bore completion. Each tool is pre-programmed to meet the operator's specific application needs and is available in a multi-cycle version. Opening and closing are activated through circulated RFID tags, pressure signatures, on-board timers, or a combination of these activation methods (Figures 20 and 21).

Figure 19—RFID Multi-Cycle Circulating Sleeve

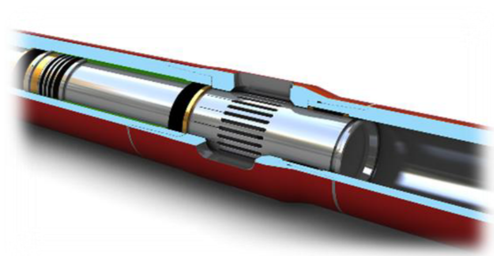

Figure 20—RFID SSD - Open Position

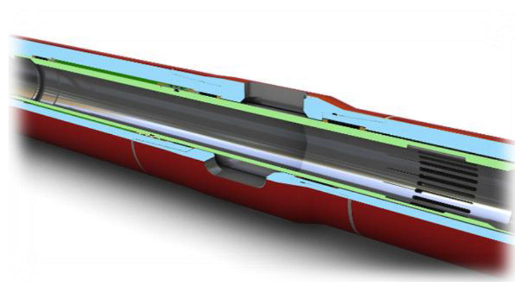

Figure 21—RFID SSD - Closed Position

The RFID Sliding Sleeve can be activated using RFID tags when a flow path is available. If a flow path is not present, activation can be achieved through a pressure-pulse method. Additionally, the onboard timer provides an alternative by enabling autonomous activation, eliminating the need for either RFID or pressure-pulse triggers.

RIFD SSD Technical Specs

Production Packer

The Retrievable production packer is a cut-to-release, and hydraulic/hydrostatic-set, that combines the durability of a permanent packer designed with the flexibility of a retrievable model. Designed as a one-trip solution, it functions effectively as a primary single-string production packer, delivering the strength of a permanent packer while allowing for straightforward retrieval when needed (Murdoch et al, 2019) (Figure 22).

Figure 22—™ Production Packer

The packer is highly adaptable, suitable for various applications including conventional, mono-bore, stacked completions, and liner-top isolations. Its robust design makes it an excellent choice for subsea installations, where packer reliability is paramount.

Engineered to surpass ISO 14310 Level V0 standards, the packer offers a high-performance envelope with exceptional load and pressure ratings, significantly reducing the risk of failure. Its simple design with minimal components enhances both reliability and operational efficiency. Additionally, the packer has been subjected to rigorous qualification testing, including hot-swab evaluations, to validate its performance in challenging environments further.

Technical Specs

Table 1—RFID RIV Technical Specs

Specifications	RFID RIV
Size	4 1/2"
Max OD	5.625"
Min ID	2.505"
Length	113"
Burst Pressure Rating	10,000psi
Collapse Pressure Rating	10,000psi
Absolute Pressure Rating	10,000psi
Temperature Rating	302ºF
Top Thread	4 1/2 VAGT 12.6# Box
Bottom Thread	4 1/2 VAGT 12.6# Pin
Tensile	353,000lbs
Elastomers	HNBR
Metallurgy	AISI 4140, 80Ksi

Table 2—RFID RIV Specification

<table>
<tr><th>Size</th><th>Maximum OD</th><th>Minimum ID</th><th>Pressure Rating</th><th>Absolute Pressure Rating</th><th>Temperature</th><th>Minimum Flow Area</th><th>Maximum Differential Opening Pressure</th></tr>
<tr><td>3.50 in. (88.9 mm)</td><td>5.85 in. (149 mm)</td><td>2.81 in. (71.4 mm)</td><td rowspan="3">7,500 psi (51.7 MPa)</td><td>15,000 psi (103.4 MPa)</td><td rowspan="3">39 to 302°F (4 to 150°C)</td><td>6.20 in.² (40 cm²)</td><td rowspan="3">1,500 psi (10.3 MPa)</td></tr>
<tr><td>5.50 in. (139.7 mm)</td><td>7.75 in. (196.9 mm)</td><td>4.31 in. (109.5 mm)</td><td>12,500 psi (86.1 MPa)</td><td>16.33 in.² (105.4 cm²)</td></tr>
<tr><td>5.50 in. (139.7 mm)</td><td>8 in. (203.2 mm)</td><td>4.56 in. (115.8 mm)</td><td>15,000 psi (103.4 MPa)</td><td>16.33 in.² (105.4 cm²)</td></tr>
</table>

Table 3—Production Packer Technical Spec

Outside Diameter (OD)		7 in. (177.8 mm)						9-5/8 in. (244.5 mm)			10-3/4 in. (273 mm)	
Weight		23 to 26 lb/ft (34.2 to 38.7 kg/m)	29 to 32 lb/ft (43.1 to 47.6 kg/m)	32 to 35 lb/ft (47.6 to 52.1 kg/m)	29 to 32 lb/ft (43.1 to 47.6 kg/m)	32 to 35 lb/ft (47.6 to 52.1 kg/m)	35 to 38 (52.1 to 56.6 kg/m)	47 to 53.5 lb/ft (69.9 to 79.6 kg/m)			60.7 to 65.7 lb/ft (90.3 to 97.7 kg/m)	
Minimum ID		6.187 in. (157.10 mm)	5.990 in. (152.10 mm)	5.892 in. (149.60 mm)	5.990 in. (152.10 mm)	5.892 in. (149.60 mm)	5.801 in. (147.35 mm)	8.405 in. (213.50 mm)			9.417 in. (239.20 mm)	
Maximum ID		6.466 in. (164.20 mm)	6.293 in. (159.80 mm)	6.208 in. (157.90 mm)	6.293 in. (159.80 mm)	6.208 in. (157.90 mm)	6.123 in. (155.52 mm)	8.822 in. (224.00 mm)			9.819 in. (249.40 mm)	
Maximum OD		6.000 in. (152.40 mm)	5.910 in. (150.10 mm)	5.820 in. (147.80 mm)	5.910 in. (150.10 mm)	5.820 in. (147.80 mm)	5.735 in. (145.67 mm)	8.310 in. (211.10 mm)			9.345 in. (237.40 mm)	
Standard thread connection		3 1/2-in. VAM® TOP			4 1/2-in. VAM® TOP		3 1/2-in. VAM® TOP	4 1/2-in. VAM® TOP	5 1/2-in. VAM® TOP HC	7-in. VAM® TOP HC	5 1/2-in. VAM® TOP HC	7-in. VAM® TOP HC
Minimum ID		2.962 in. (75.23 mm)			3.833 in. (97.40 mm)		2.720 in. (69.09 mm)	3.894 in. (98.91 mm)	4.650 in. (118.10 mm)	6.060 in. (149.20 mm)	4.750 in. (120.60 mm)	6.060 in. (149.20 mm)
80-ksi material	Differential pressure rating	10,000 psi (69 MPa)			8,400 psi (58 MPa)		N/A	8,000 psi (55 MPa)		6,700 psi (46 MPa)	6,500 psi (45 MPa)	
	Tensile rating	200,000 lb (890 KN)			178,128 lb (792 KN)		N/A	400,000 lb (1,779 KN)	360,380 lb (1,603 KN)	302,264 lb (1,345 KN)	400,000 lb (1,779 KN)	
95-ksi material	Differential pressure rating	10,000 psi (69 MPa)					15,000 psi (103 MPa)	8,000 psi (55 MPa)			6,500 psi (45 MPa)	
	Tensile rating	200,000 lb (890 KN)			212,546 lb (945 KN)		212,000 lb (943 KN)	400,000 lb (1,779 KN)		360,380 lb (1,603 KN)	400,000 lb (1,779 KN)	
110-ksi material	Differential pressure rating	10,000 psi (69 MPa)					15,000 psi (103 MPa)	8,000 psi (55 MPa)			6,500 psi (45 MPa)	
	Tensile rating	200,000 lb (890 KN)			212,546 lb (945 KN)		212,000 lb (943 KN)	400,000 lb (1,779 KN)				
Element type		HNBR			HNBR	HNBR & AFLAS	AFLAS	HNBR				
Flow-test rating at 80°F (27°C)		N/A			10 bbl/min (1.6 m³/min)		N/A	14 bbl/min (2.2 m³/min)				
Flow-test rating at 176°F (79°C)		N/A						10 bbl/min (1.6 m³/min)				
Temperature range		80 to 350°F (27 to 177°C)					200 to 400°F (93 to 204°C)	80 to 350°F (27 to 177°C)				
ISO 14310-tested		V0										

Production Packer PBR

The Polished Seal Bore Receptacle is engineered for completions that require maintaining a large internal diameter while managing significant tubing movement.

Its modular design enables easy conversion between tension release and hydraulic release configurations. Both versions feature a polished seal bore receptacle and a PBR seal assembly that includes four sets of working seals and debris barriers positioned above and below them. When used with a hydraulic or hydraulic/hydrostatic setting mechanism, the system can be deployed and set in a single trip. The seal assembly is retrieved along with the tubing string, while the PBR bore is removed using a dedicated retrieval tool, typically in a second trip (Figure 23).

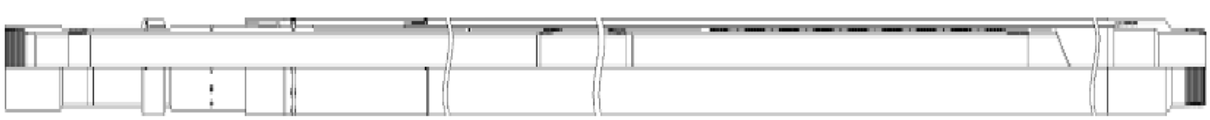

Figure 23—Polished Seal Bore Receptacle

The receptacle is available in various lengths, offering stroke capabilities of up to 20 feet. An adjustable shear mechanism secures the outer housing to the seal mandrel during installation. Once released, the seal assembly is free to move within the bore to accommodate tubing expansion or contraction. For one-trip completions, the PBR assembly can be pinned in fully stroked, closed, or intermediate positions to meet specific spacing requirements.

Technology Trial Planning & Risk Assessment

A thorough risk assessment (as a planning process) has to be conducted to ensure all potential scenarios are considered and a suitable contingency measure is in place for each. The conventional ball seat isolation valve could be added as a contingency option in case the RFID tag did not close the RIV as intended. Close co-operation and planning with all parties are required before the RFID-RIV technology can be deployed and tested on the well. The objective of the test is to qualify the RIV to close via the tag command, with a 100hours / 4 day back-up timer set in case the lower completion could not be isolated via RFID tags or the conventional ball drop system. This, in effect, provided three layers of actuation options to ensure the completion could be isolated.

Primary Method > RFID Tags to close RIV
Secondary Method > Conventional Ball-drop - Seat System
Tertiary Method > Electronic Timer on RIV

Conclusion

The lower completion system, comprising the RFID RIV and RFID Opti-Ross SSD, was successfully installed in different countries worldwide, achieving all operational objectives. Plans are underway for its deployment in the Middle East. A key design feature is its enhanced flow dynamics for efficient and predictable toe isolation. This is realized through an RFID tag, housed in a tag carrier, which demonstrates superior conformance to fluid velocity compared to traditional ball-drop methods susceptible to trajectory-induced velocity deviations.

Contingency and risk are substantially minimized by an integrated backup electronic timer, providing a safeguard against unfavorable reservoir conditions or complete loss of circulation during the completion phase. These novel methodologies significantly reduce operational risk, time, and expenditure, while also offering the capability to extend horizontal wellbores in specific global regions. Moreover, the RFID multi-cycle circulating sliding sleeve permits intervention-less circulation of production fluids.

Acknowledgments

The authors extend their sincere gratitude to the entire team, whose dedication was instrumental in the development and successful global deployment of the RFID tools across diverse applications. These successful field operations have established a significant and positive operational track record for this cutting-edge technology.

Nomenclature

RFID– Radio Frequency Identification Device
RIV– Reservoir Isolation Valve
PBR– Polished Bore Receptacle
RIH– Run in Hole
ICD– Inflow Control Device
CT– Coiled Tubing
ISO– International Organization for Standardization

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